

PROJECT REPORT: AI Resume Analyzer (Database System)

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1. Introduction

The Resume Analyzer System is a database-based project designed to store, manage, and analyze user resumes by comparing their skills with job requirements. The system helps in identifying suitable candidates for specific job roles based on their skills and experience.

This project uses SQL (Structured Query Language) to create and manage the database.

2. Objectives

- To store user resume data efficiently
- To manage skills and job requirements
- To match users with suitable jobs
- To demonstrate database relationships (ER Model)
- To perform queries for analysis

3. Tools & Technologies

- Database: MySQL
- Language: SQL
- Platform: MySQL Workbench / XAMPP

4. Database Structure

Users Table:

- user_id (Primary Key)
- name
- email
- education
- experience_years

Skills Table:

- skill_id (Primary Key)
- skill_name

UserSkills Table:

- user_id (Foreign Key)
- skill_id (Foreign Key)
- proficiency_level

Companies Table:

- company_id (Primary Key)
- company_name

Jobs Table:

- job_id (Primary Key)
- company_id (Foreign Key)
- job_title

JobSkills Table:

- job_id (Foreign Key)
- skill_id (Foreign Key)

5. Relationships

- One User → Many Skills
- One Job → Many Skills
- Many Users ↔ Many Skills
- Many Jobs ↔ Many Skills
- One Company → Many Jobs

6. Data Description

- 100 Users inserted
- 20 Skills inserted
- Multiple companies and jobs

7. Key Functionalities

- Store user data
- Skill mapping
- Job matching
- Query-based analysis

8. Sample Queries

Find users with SQL skill:

```
SELECT u.name  
FROM Users u  
JOIN UserSkills us ON u.user_id = us.user_id  
JOIN Skills s ON us.skill_id = s.skill_id  
WHERE s.skill_name = 'SQL';
```

Find jobs requiring Python:

```
SELECT j.job_title  
FROM Jobs j  
JOIN JobSkills js ON j.job_id = js.job_id  
JOIN Skills s ON js.skill_id = s.skill_id  
WHERE s.skill_name = 'Python';
```

9. Advantages

- Organized data management
- Easy job matching
- Scalable design

10. Limitations

- No frontend
- No real AI
- Manual data

11. Future Improvements

- AI-based resume scoring
- Web interface
- Resume parsing

12. Conclusion

This project shows how databases help in resume management and job matching.